Hamro Hospital
Updated HIS Workflow Integration Report

Doctor, Referral, Department, Billing, Cash Counter, Admission, Nursing, OT, Blood Bank, Lab, Radiology, Pharmacy and Finance workflow — with remaining recommendations.

Important clarification

The latest workflow implementation did not intentionally redesign the main frontend/backend UI. The existing cards, sidebar, dashboards, tables, buttons and Jazzmin admin layout were kept. Functional additions were added inside the existing style. Print/receipt documents are the only UI area intentionally improved because hospital logo, patient barcode, bill barcode and professional print layout were required.

What this report gives

• Detailed explanation of current connections made inside the project.

• Simple figures showing the system flow.

• Clear list of what is working now.

• Clear list of what is still missing or recommended.

• My suggested architecture logic to make this a best-practice hospital system.

1. Current Project Connection Summary

Doctor → Referral → Dept Queue → Bill → Cashier → Paid → Service

The current system is connected around a practical operational path: doctor searches a patient, creates referral/order, destination department is notified, department reviews request, department generates pending bill, Cash Counter collects payment, payment status becomes Paid, and department continues service.

Module	Connection Made
Doctor	Search patient, view patient records, create referrals, view own referrals, update profile.
Referral	Carries diagnosis, reason, clinical notes, instructions, attachment, requested items and related bill.
Notifications	Role/user notifications with related links to referral detail.
Billing	Pending/Paid workflow, receipt status, patient/bill barcode, selected bill payment.
Cash Counter	Pending bills visible, scan/search by patient or bill, pay selected bills.
Departments	Incoming referrals visible in Lab/Radiology/Pharmacy dashboards; same pattern can be expanded.
Admission	Admission referrals, ward daily rate, estimated charge, pending bill discharge block.
Patient Records	Doctor can view record sections, documents view-only, previous referrals added.

2. UI/UX Protection Statement

Your concern is valid: a hospital project should not randomly change UI each time functionality is added. The latest implementation uses existing design components and does not replace the main frontend/backend UI. Any visible additions are new features inserted into existing pages.

Area	Changed?	Reason
Main dashboard UI	No redesign	Existing dashboard_base structure retained.
Admin/Jazzmin	No design change	Only backend admin save/link behavior was fixed.
Doctor pages	Feature additions	Search, profile, referral links added using same layout.
Department dashboards	Feature additions	Incoming referral panels added using existing card/table style.
Billing receipts	Yes, print-focused	Mandatory logo, patient barcode, bill barcode and status were requested.
Website public UI	Earlier separate changes	Only because previously requested: navbar/home notification etc.

Rule going forward

Keep UI locked. Only add functional blocks where required. For any print document, use a single print identity: hospital logo, hospital name, patient details, patient barcode, module/bill barcode, date/time, status, prepared/printed by.

3. Doctor to Patient Record Workflow

Search → Patient → Record Tabs → View Only → Refer

Doctors can search patient by Hospital ID/barcode text, phone, or name. The doctor can then open the patient profile and access sections including billing, insurance, admission, laboratory, radiology, blood bank, pharmacy, medical reports, appointments, visits, medical records/documents and referrals.

Patient Area	Doctor Permission
Profile	View
Patient card/barcode	View only
Billing/receipts	View
Lab/Radiology reports	View
Pharmacy/admission/blood bank	View history
Medical documents	View only
Download documents	No, except Super Admin/Registration
Print/download patient card	No, except Super Admin/Registration
Create referrals	Yes

4. Doctor Account and Profile Logic

Doctor accounts are now tied to hospital staff users. When a doctor profile is created from the Super Admin doctor form or Django Admin and no user account is linked, the system creates a Doctor role user and links it to the doctor profile.

Problem	Implemented Logic
Doctor cannot log in	Auto-create/link User account with role Doctor.
Doctor dashboard incomplete	Added dashboard, patient search, referral, profile and password links.
Profile warning about not linked	Reduced by auto-linking active doctors; optional setup messages only remain for genuinely incomplete p
eSewa warning confusion	eSewa fields are optional; no forced warning when blank.
Doctor as patient	Doctor User and Patient record remain separate and can coexist.

Recommendation

Add an explicit staff onboarding page where Super Admin can generate username/password, force password reset on first login, and optionally send credentials by SMS/email.

5. Referral Category Logic

Instead of one long scrolling referral form, the workflow now supports category selection first. After the doctor chooses Laboratory, Radiology, Pharmacy, Admission, Nursing, OT, Blood Bank or another department, the form shows only the relevant structured options plus common notes/attachment fields.

Referral Type	Data Captured
Laboratory	LabTest catalogue checkboxes and clinical notes.
Radiology	RadiologyTest catalogue checkboxes and instructions.
OT	OperationType/procedure choices and notes.
Blood Bank	Blood group, units, urgency, crossmatch/product details in request text.
Admission	Admission notes, expected ward/care information.
Nursing	Nursing instructions such as IV fluid, dressing, monitoring.
Pharmacy	Medicine/requested items, notes, attachment.
Other doctor/department	General clinical referral to destination department doctors.

6. Department Notification and Queue Logic

When a referral is saved, a notification is generated. For fixed departments, notification is role-based. For doctor-to-doctor or department referral, doctors linked to the destination department are notified.

Destination	Notification Target
Laboratory	Laboratory role users
Radiology	Radiology role users
Pharmacy	Pharmacy role users
Admission	Ward/Admission role users
Nursing	Nursing role users
OT	Operation Theatre role users
Blood Bank	Blood Bank role users
Clinical department	Doctors in that department

Current limitation

Notifications are database notifications displayed in the navbar/sidebar after refresh. True live notification requires WebSockets/Django Channels.

7. Department → Pending Bill → Cash Counter

Dept Accept → Generate Bill → Pending → Patient Pays → Paid → Dept Notified

Departments can generate a pending bill from a referral. The bill is linked back to the referral. The printed receipt displays Pending status and payment instructions. Cash Counter then collects payment and marks the bill Paid.

Payment Step	System Behavior
Department generates bill	Bill status Pending; referral.related_bill set.
Receipt printed	Shows Pending, patient barcode, bill barcode.
Cashier scans/searches	Pending bills are shown by patient ID/phone/bill number.
Cashier pays selected	Bill status changes to Paid, cashier stored.
Department update	Payment Completed notification created.
Finance	Paid bills become visible in reports/revenue.

8. Cash Counter and Daily Queue

Cash Counter shows today's paid activity and active pending payments. Historical data is not deleted; it remains searchable through reports and filters. This matches the requested daily reset principle: dashboard queue resets operationally, database remains permanent.

Cashier Requirement	Status
Scan patient barcode	Supported by patient lookup / scanner widget and patient ID lookup.
Scan bill barcode	Bill number can be searched; bill barcode encodes bill number.
Show pending bills	Pending Payments section and patient lookup pending-bill block added.
One-click payment	Pay Selected supports multiple pending bills.
Print receipt	Receipt remains available before and after payment.
Department notified after payment	Implemented through related referral notifications.

9. Department Workflows Connected

Department	Current Connection
Laboratory	Incoming referrals visible; structured lab tests; pending bill; payment status visible.
Radiology	Incoming referrals visible; structured radiology services; pending bill/payment workflow.
Pharmacy	Incoming referrals visible; requested medicines/items; can follow pending payment policy.
Admission	Admission referrals visible; ward/bed allocation; rates and discharge payment check.
Nursing	Receives referral instructions; can view patient and referral. Structured nursing requests still recommended.
OT	Procedure referrals and OT billing path; richer OT booking form still recommended.
Blood Bank	Blood bank referral route exists; detailed structured form still recommended.
Finance	Paid bills feed finance/revenue reports; department-specific revenue should be expanded further.

10. Admission, Ward, Bed and Discharge

Doctors cannot admit directly. Admission staff review admission referrals, admit patient, assign ward/bed, and handle discharge. Ward has a configurable daily rate and admission slip shows estimated charge.

Admission Function	Status
Barcode/Hospital ID/phone search	Admission search supports patient ID/name/phone; scanner widget available.
Bed availability	Ward/bed page shows Available green and Occupied red.
Daily ward rate	Ward.daily_rate added.
Charge calculation	Length of stay × daily rate estimate added.
Admission revenue dashboard	Initial admission revenue metrics added from paid IPD bills.
Discharge block	Discharge blocked if pending bills exist.
Full discharge package	Recommended next step.

11. Print and Receipt Standard

Print documents are the only area where visual improvement was intended. Standard hospital receipt logic should include hospital identity, patient identity, patient barcode, document/bill barcode, service name, amount, status, printed by and printed date/time.

Document	Current Print Standard
Billing receipt	Patient barcode, bill barcode, payment status, hospital identity.
OPD ticket	Patient phone, patient barcode/QR, visit barcode, payment status.
Admission slip	Hospital identity, patient barcode, admission barcode, charge estimate.
Lab/Radiology reports	Hospital identity, patient barcode and module barcode.
Pharmacy receipt	Hospital identity, patient barcode where patient exists, sale barcode.
OT/Blood/Insurance	Barcodes and patient identity included in print flow where applicable.

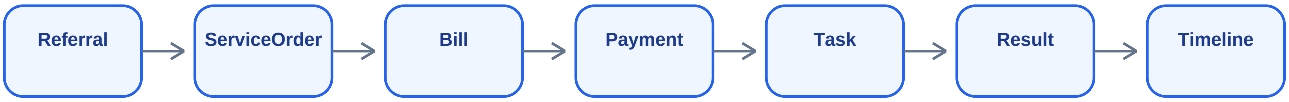
Recommendation

Build one reusable print partial/component so every future receipt automatically has the same header, barcode block, footer and print actions. This will prevent inconsistency.

12. What Is Missing / Still Needed

Priority	Missing / Recommended	Why It Matters
High	Central ServiceOrder engine	Prevents duplicate logic across referral, lab, radiology, pharmacy, OT and admission.
High	Real-time WebSocket notifications	Departments should see new orders/payment completion without refreshing.
High	Full eSewa payment for pending bills	Online payment should mark pending bills paid automatically.
High	Structured Blood Bank form	Blood group, product, crossmatch, units and urgency need proper fields.
High	Structured Nursing request form	Admission/transfer/discharge readiness should be workflow objects.
High	Full discharge checklist	Blocks discharge until payments, reports and documents are complete.
Medium	Unified receipt template	Ensures every printable document remains consistent.
Medium	Pagination and filters	Large hospital datasets will slow without pagination.
Medium	Patient timeline	One chronological view of visits, referrals, bills, reports and admissions.
Medium	Department-specific revenue dashboards	Department users should see own revenue; Finance sees all.
Medium	Pharmacy prescription-to-sale	Doctor prescription should convert directly to pharmacy dispensing queue.
Medium	Lab/Radiology result notification to doctor	Doctors should be notified when result is completed.
Low	Discharge package generation	Final PDF bundle for patient record and discharge handover.

13. My Recommended Core Logic



The next best architectural step is a central ServiceOrder model. Referrals should create one or more ServiceOrders. Each ServiceOrder can be billable or non-billable. Bills can cover one or many orders. Department dashboards display orders, not scattered model-specific requests.

Proposed Model	Core Fields
ServiceOrder	patient, referral, ordered_by, department_type, service_name, price, status, bill
PaymentEvent	bill, amount, method, received_by, paid_at, gateway_reference
DepartmentTask	service_order, assigned_role, accepted_by, completed_by, result_document
PatientTimeline	patient, event_type, source_object, timestamp, summary
Notification	role/user, title, message, related_url, read_at

14. Final Status and Roadmap

Area	Current Status
Doctor patient history view	Implemented with sections and view-only document permissions.
Patient card permissions	Implemented: doctors view only; print/download for Super Admin and Registration.
Doctor profile	Implemented with optional eSewa fields.
Referral category flow	Implemented.
Structured Lab/Radiology/OT selection	Implemented using catalogues.
Department notifications	Implemented as database notifications with links.
Pending payment workflow	Implemented.
Cashier pay selected	Implemented.
Admission rates and discharge block	Implemented initial version.
Enterprise order engine	Recommended next.
Live real-time updates	Recommended next.
Final perfect HIS	Not complete yet; strong working foundation exists.

Best next step

Build the Central ServiceOrder module first. It will make every future feature easier: one queue format, one payment lifecycle, one report system, one patient timeline, one department task system. After that, add WebSocket live notifications and eSewa pending-bill completion.

15. Suggested Implementation Roadmap

Phase	Work	Expected Benefit
Phase 1	Create ServiceOrder model and migrate referral bill logic to orders.	Unified HIS backbone.
Phase 2	Build department order queue component reused by Lab, Radiology, Pharmacy, OT, Blood Bank, Nursing.	Less duplicate code, consistent workflow.
Phase 3	Add PaymentEvent and eSewa callback for pending bills.	Online payment works same as cashier payment.
Phase 4	Add PatientTimeline page.	Complete chronological patient history.
Phase 5	Add WebSockets/Django Channels.	Real-time department notifications.
Phase 6	Create unified print template partial.	Every receipt/slip/report stays consistent.
Phase 7	Add discharge checklist and package.	Safer admission/discharge workflow.
Phase 8	Add analytics and role-specific revenue dashboards.	Finance and department leadership reports.
Phase 9	Add audit/permission hardening and tests.	Production readiness.
Phase 10	Performance: pagination, indexes, caching for large data.	Scales to real hospital workload.

Conclusion

The current system now has a connected workflow foundation. To make it best, do not keep adding isolated features. Build the central order/payment/timeline architecture, then all departments will become easier to maintain and more realistic.